30/09/2026

CODE-B
Phase-2

Aakash
Medical | IIT-JEE | Foundations

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FORTNIGHTLY TEST FOR NEET-2027 (XII PASSED)

MM: 720 **Test - 05** **Time: 180 Mins.**

Answers

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1.	(1)	37.	(1)	73.	(1)	109.	(2)	145.	(4)
2.	(3)	38.	(2)	74.	(1)	110.	(3)	146.	(3)
3.	(1)	39.	(1)	75.	(3)	111.	(3)	147.	(1)
4.	(3)	40.	(2)	76.	(3)	112.	(3)	148.	(4)
5.	(3)	41.	(4)	77.	(1)	113.	(2)	149.	(1)
6.	(1)	42.	(3)	78.	(1)	114.	(3)	150.	(2)
7.	(4)	43.	(3)	79.	(4)	115.	(1)	151.	(3)
8.	(2)	44.	(4)	80.	(2)	116.	(1)	152.	(2)
9.	(2)	45.	(4)	81.	(1)	117.	(2)	153.	(3)
10.	(2)	46.	(2)	82.	(2)	118.	(3)	154.	(1)
11.	(2)	47.	(4)	83.	(3)	119.	(2)	155.	(2)
12.	(1)	48.	(3)	84.	(3)	120.	(2)	156.	(1)
13.	(1)	49.	(2)	85.	(2)	121.	(3)	157.	(3)
14.	(1)	50.	(3)	86.	(4)	122.	(3)	158.	(3)
15.	(2)	51.	(2)	87.	(4)	123.	(4)	159.	(4)
16.	(2)	52.	(2)	88.	(2)	124.	(3)	160.	(1)
17.	(4)	53.	(1)	89.	(3)	125.	(4)	161.	(4)
18.	(1)	54.	(2)	90.	(1)	126.	(2)	162.	(1)
19.	(3)	55.	(2)	91.	(3)	127.	(1)	163.	(2)
20.	(2)	56.	(3)	92.	(4)	128.	(2)	164.	(2)
21.	(4)	57.	(2)	93.	(3)	129.	(2)	165.	(3)
22.	(3)	58.	(1)	94.	(1)	130.	(2)	166.	(3)
23.	(1)	59.	(2)	95.	(4)	131.	(3)	167.	(3)
24.	(2)	60.	(4)	96.	(2)	132.	(3)	168.	(1)
25.	(4)	61.	(2)	97.	(3)	133.	(4)	169.	(1)
26.	(2)	62.	(1)	98.	(3)	134.	(2)	170.	(3)
27.	(2)	63.	(1)	99.	(2)	135.	(2)	171.	(3)
28.	(3)	64.	(2)	100.	(1)	136.	(3)	172.	(1)
29.	(1)	65.	(3)	101.	(1)	137.	(2)	173.	(3)
30.	(2)	66.	(3)	102.	(2)	138.	(2)	174.	(4)
31.	(3)	67.	(1)	103.	(2)	139.	(1)	175.	(2)
32.	(2)	68.	(2)	104.	(1)	140.	(4)	176.	(1)
33.	(1)	69.	(3)	105.	(3)	141.	(1)	177.	(2)
34.	(4)	70.	(2)	106.	(1)	142.	(3)	178.	(4)
35.	(1)	71.	(2)	107.	(1)	143.	(3)	179.	(2)
36.	(2)	72.	(2)	108.	(3)	144.	(1)	180.	(3)

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Hints and Solutions

PHYSICS

(1) **Answer :** (1)
Solution:
Temperature is the measure of hotness and coldness.

(2) **Answer :** (3)
Solution:
$\frac{T_C-0}{100-0}=\frac{T-10}{130-10}$
$\frac{40-0}{100-0}=\frac{T-10}{130-10}$ or $\frac{40}{100}=\frac{T-10}{120}$
$\Rightarrow \frac{2}{5}=\frac{T-10}{120}$
$\Rightarrow 5T - 50 = 240$ or $T=\frac{290}{5}=58^\circ$

(3) **Answer :** (1)
Solution:
At thermal equilibrium two bodies are at same temperature.

(4) **Answer :** (3)
Solution:
Using $\frac{\Delta T}{T_0}=\frac{\alpha\Delta\theta}{2}$
$=\frac{2\times10^{-3}\times50}{2}=50\times10^{-3}=0.05$

(5) **Answer :** (3)
Solution:
For an isotropic solid $\Rightarrow \gamma = 3\alpha$

(6) **Answer :** (1)
Solution:
$V=V_0\,(1+\gamma\Delta\theta)$
V_0 = initial volume
$\Delta\theta$ = temperature difference
γ = coefficient of volumetric expansion
Since, V_0, γ and $\Delta\theta$ are same for both the spheres, therefore they will expand by equal amount.

(7) **Answer :** (4)
Solution:
$Q=mc_{ice}\Delta T+mL+mc_w\Delta T$
= 10 × 0.5 × 10 + 10 × 80 + 10 × 1 × 20
= 50 + 800 + 200 = 1050 cal

(8) **Answer :** (2)
Hint:
Heat capacity $H = mC$ (C : specific heat capacity)
Solution:
$\frac{R_1}{R_2}=\frac{1}{3}$ and $\frac{C_1}{C_2}=\frac{1}{1}$
Hence capacity (H) = mC
$\frac{H_1}{H_2}=\frac{m_1C_1}{m_2C_2}=\frac{\frac{4}{3}\pi{R_1}^3.C}{\frac{4}{3}\pi{R_2}^3.C}$
$=\left(\frac{R_1}{R_2}\right)^3=\frac{1}{27}$

(9) **Answer :** (2)
Solution:
Change on kelvin scale in same as Celsius scale as number of division on both scales are same

(10) **Answer :** (2)

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Solution: According to Newton's law of cooling, Rate of cooling is directly proportional to temperature difference between the body and the surrounding. So, as the temperature difference decreases rate of cooling will decrease which means it will take more time cool down or we can say that it's temperature drop will be less during the same interval. In the first 10 minutes, temperature drops by 20 degrees, so in next 10 minutes it will drop by less than 20 degrees

(11) **Answer :** (2)
Solution:
Let the temperature of junction is T.
$\frac{60-T}{R} = \frac{T-0}{R} + \frac{T-30}{R}$
T = 30°C $\Rightarrow$ No heat flow from C

(12) **Answer :** (1)
Hint:
$W = P\Delta V = \Delta Q - \Delta U$
Solution:
$W = P\Delta V = nR\Delta T$ for isobaric process
$W = -\Delta U$ for adiabatic process
$P\Delta V = 0$ for isochoric process
$W = \Delta Q$ for Isothermal process

(13) **Answer :** (1)
Solution:
Volume in both paths (1) and (2) is increasing, so work done in both paths is also increasing.

(14) **Answer :** (1)
Solution:
$\because E = \frac{fRT}{2}$
i.e. $E \propto f$
$\frac{E_1}{E_2} = \frac{f_{Ne}}{f_{H_2}} = \frac{3}{5}$

(15) **Answer :** (2)
Solution:
$W_{ABCD} = W_{AB} + W_{BC} + W_{CD}$
$= P_0V_0 + 0 - 2P_0V_0$
$= -P_0V_0$

(16) **Answer :** (2)
Hint:
For a process having equation of state as
PV^X = Constant
Work done $W = \frac{nR\Delta T}{1-x}$
Solution:
Here,
$\frac{P}{\sqrt{V}}$ = Constant
$\Rightarrow PV^{-\frac{1}{2}}$ = Constant
Now,
$W = \frac{nR\Delta T}{1-x} = R\left[\frac{30}{1-\left(-\frac{1}{2}\right)}\right]$
$W = 20R$

(17) **Answer :** (4)
Solution:
Internal energy is a path fuction. therefore for closed path change in internal energy must be zero.
Ans: D

(18) **Answer :** (1)
Hint:
Use first law of thermodynamics
Solution:
$\Delta Q = \Delta U + \Delta w$
$\Rightarrow \Delta U = \Delta Q - \Delta w$

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$\Rightarrow \Delta U = 150 - 80 = 70$ J

(19) **Answer :** (3)
Solution:
$C = \frac{Q}{m\Delta T}$
$\Delta T = 0, \quad C \to \infty$

(20) **Answer :** (2)
Solution:
$\because PV = nRT$
$\Rightarrow P = \left(\frac{nR}{V}\right) T$
$\frac{nR}{V}$ is slope of $P - T$ graphs
$\therefore V_1 < V_2$

(21) **Answer :** (4)
Solution:
Internal energy change
$\Delta U = nC_V \Delta T$ depends on gas nature but not on process.
and temperature change

(22) **Answer :** (3)
Hint:
For adiabatic process PV^γ = constant
Solution:
$PV = nRT$
$V = \frac{nRT}{P}$
PV^γ = constant
$P\left(\frac{nRT}{P}\right)^\gamma$ = constant
$P \propto (T)^{\frac{\gamma}{\gamma-1}}$
$\Rightarrow \frac{\gamma}{\gamma-1} = k$
and $\gamma = \frac{7}{5}$
$k = \frac{7}{2}$

(23) **Answer :** (1)
Solution:
For isochoric process, $P \propto T$

(24) **Answer :** (2)
Solution:
Mean kinetic energy of a molecule per degree of freedom is $\frac{1}{2}k_BT$ and gas molecules have three translational degrees of freedom.

(25) **Answer :** (4)
Solution:
Maximum efficiency = $1 - \frac{T_1}{T_2} = 1 - \left(\frac{400}{600}\right)$ = 33.33%

(26) **Answer :** (2)
Solution:
Irreversible engine has less efficiency due to the dissipative effects while a reversible engine consists of reversible processes which are quasi-static and non-dissipative.

(27) **Answer :** (2)
Solution:
$\Delta Q = 0$ in adiabatic process,
$WD + \Delta U = 0$
$-\Delta U = WD$

(28) **Answer :** (3)
Solution:
A carnot cycle has 2 reversible adiabatic processes and 2 reversible isothermal processes.

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(29) **Answer :** (1)
Hint:
Use, $\frac{C_P}{C_V} = \gamma$
Solution:
$\gamma = \frac{f+2}{f}$
Degree of freedom for a monatomic gas is 3, and for a diatomic gas is 5.
$\gamma_{mono} = \frac{5}{3}$
$\gamma_{dia} = \frac{7}{5}$

(30) **Answer :** (2)
Solution:
$v_{rms} = \sqrt{3RT/M}$
The root-mean-square speed of an ideal gas at 800 K is exactly twice its value at 200 K

(31) **Answer :** (3)
Solution:
$\lambda_m = \frac{1}{\sqrt{2}\pi\, nd^2}$ ...(i)
and $PV = NKT$
$P = \left(\frac{N}{V}\right)KT$
$P = nKT$
from (i) and (ii)
$n = \frac{P}{KT}$ ...(ii)
$\lambda_m = \frac{KT}{\sqrt{2}\pi Pd^2}$
$\lambda_m \propto \frac{1}{P}$ $\left[\because T = \text{constant}\right]$
If $P \rightarrow 2P$
$\lambda_m' = \frac{\lambda_m}{2}$

(32) **Answer :** (2)
Solution:
$P_2 = P_0 \left[\frac{V_1}{V_2}\right]^{\gamma} = 32\ P_0$

(33) **Answer :** (1)
Solution:
Argon being monoatomic has only translational degree of freedom.

(34) **Answer :** (4)
Solution:

$$U = n_1\frac{f_1}{2}RT + n_2\frac{f_2}{2}RT$$
$$= 2\times\frac{5}{2}RT + 4\times\frac{3}{2}RT$$
$$= 5RT + 6RT$$
$$U = 11RT$$

(35) **Answer :** (1)
Solution:
For ideal gas,
$PV = nRT$
$V = \text{constant}$
$P \propto T$
$\therefore \frac{\Delta P}{P} = \frac{\Delta T}{T}$
$\frac{1}{100} \times 0.5 = \frac{\Delta T}{T}$
$\Rightarrow T = \frac{4\times 1000}{5}$
$T = 800\ \text{K} = 527°\text{C}$

(36) **Answer :** (2)

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Solution:

Most probable speed = $\sqrt{\frac{2RT}{M}}$

(37) Answer : (1)
Solution:

$\frac{\rho}{P} \propto \frac{1}{T}$

$\frac{\left(\frac{\rho}{P}\right)_{10^\circ C}}{\left(\frac{\rho}{P}\right)_{110^\circ C}} = \frac{383}{283} \Rightarrow \frac{5}{\left(\frac{\rho}{P}\right)_{110^\circ C}} = \frac{383}{283}$

$\therefore \left(\frac{\rho}{P}\right)_{110^\circ C} = \frac{283}{383} \times 5 = 3.69$

(38) Answer : (2)
Solution:

For isochoric process the volume is constant

$\frac{\Delta P}{\Delta V} = \tan\theta = \text{slope}$

$\Delta V = 0$

Slope = ∞

(39) Answer : (1)
Solution:

Air trapped between double walls acts as an insulator.

(40) Answer : (2)
Solution:

Specific heat $C = \frac{\mathrm{J}}{\mathrm{Kg\ K}}$

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Heat capacity $C = \frac{\mathrm{J}}{\mathrm{K}}$

Latent heat $L = \frac{\mathrm{J}}{\mathrm{Kg}}$

Coefficient of linear expansion (α) = K^{-1}

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(41) Answer : (4)
Solution:

Natural convection requires gravity but forced convection can occur in gravity or gravity free space.

(42) Answer : (3)
Solution:

From 0°C to 4°C, volume of water decrease, so density increase and become maximum at 4°C.

(43) Answer : (3)
Solution:

$\frac{\mathrm{dQ}}{\mathrm{dt}} = \mathrm{KA}\frac{\Delta T}{\Delta x}$ or $\mathrm{Wm^{-1}\ K^{-1}}$

(44) Answer : (4)
Hint:

$\lambda_m \propto \frac{1}{T}$

Solution:

$\frac{\lambda_{m_1}}{\lambda_{m_2}} = \frac{T_2}{T_1}$

$\lambda_{m_2} = \frac{5000 \times 4000}{2000}$ = 10000 Å

(45) Answer : (4)
Solution:

$\lambda = \frac{1}{\sqrt{2}\cdot\pi n d^2} = \frac{K_B T}{\sqrt{2}\pi d^2 P}$

CHEMISTRY

(46) Answer : (2)
Solution:

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Equilibrium is dynamic in nature that is both forward and backward processes take place with same rate.

(47) **Answer :** (4)
Solution:
The equilibrium constant does not depend upon initial concentrations and has a definite value for every chemical reaction at a given temperature.

(48) **Answer :** (3)
Hint:
Multiply 3 in equation (3) and reverse equation (1) and add both these equations in equation (2)
Solution:

$2\,NH_3$	$\rightleftharpoons$	$N_2 + 3H_2$	$\frac{1}{K_1}$	$\ldots(1)$
$N_2 + O_2$	$\rightleftharpoons$	$2\,NO$	K_2	$\ldots(2)$
$3H_2 + \frac{3}{2}O_2$	$\rightleftharpoons$	$3H_2O$	K_3^3	$\ldots(3)$
$2\,NH_3 + \frac{5}{2}O_2$	$\rightleftharpoons$	$2NO + 3H_2O$	K	

$\therefore\ K = \frac{K_2 K_3^3}{K_1}$

(49) **Answer :** (2)
Hint:

$K_p = \frac{\left(P_{N_2}\right)\left(P_{H_2}\right)^3}{\left(P_{NH_3}\right)^2}$

Solution:

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$$

eq. $\quad 1-x \qquad \frac{x}{2} \qquad \frac{3x}{2}\ ,\ P_T$

Total moles at equilibrium = $1 - x + \frac{x}{2} + \frac{3x}{2} = 1 + x$

$p_{NH_3} = \left(\frac{1-x}{1+x}\right)P_T,\quad p_{N_2} = \left(\frac{x}{2(1+x)}\right)P_T$

$p_{H_2} = \left(\frac{3x}{2(1+x)}\right)P_T$

At eqbm,

$K_p = \frac{(p_{N_2}).(p_{H_2})^3}{(p_{NH_3})^2} = \frac{\left(\frac{x}{2}\right)^1\left(\frac{3x}{2}\right)^3}{(1-x)^2}.\left(\frac{P_T}{1+x}\right)^{4-2}$

$= \frac{27x^4P_T^2}{16}$

$[\because x << 1\ \therefore 1 - x = 1 + x \approx 1]$

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(50) **Answer :** (3)
Solution:
In heterogeneous equilibrium phase of components should be different.

(51) **Answer :** (2)
Hint:

$A(s) \rightarrow \underset{p_1}{B(g)} + \underset{p_1+p_2}{C(g)}$

$D(s) \rightarrow \underset{p_1+p_2}{C(g)} + \underset{p_2}{E(g)}$

Solution:

$K_{p_1} = (p_1)\,(p_1 + p_2)$

$a = p_1^2 + p_1p_2$ ...(i)

$K_{p_2} = (p_2)\,(p_1 + p_2)$

$b = p_2^2 + p_1p_2$ ...(ii)

$a + b = p_1^2 + 2p_1p_2 + p_2^2$

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$(a + b) = (p_1 + p_2)^2$ ...(iii)

Calculating total pressure:

$p_T = p_B + p_C + p_E$

$= p_1 + p_1 + p_2 + p_2$

$= 2p_1 + 2p_2$

$= 2(p_1 + p_2)$

Putting the value of $p_1 + p_2$ in equation (iii)

$p_T = 2\left(\sqrt{a + b}\right) atm$

(52) Answer : (2)

Solution:

If $\Delta G° < 0$, then $-\frac{\Delta G°}{RT}$ is positive and $e^{-\frac{\Delta G°}{RT}} > 1$, making $K > 1$.

(53) Answer : (1)

Solution:

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

1	0	0
1 – 0.2	0.2	0.1

$= 0.8$

$K_C = \frac{0.1 \times 0.2 \times 0.2}{0.8 \times 0.8}$

$K_C = 6.25 \times 10^{-3}$

(54) Answer : (2)

Solution:

$CH_4(g) + H_2O(g) \longrightarrow CO(g) + 3H_2(g)$

		a	3a

$CO(g) + 2H_2(g) \longrightarrow CH_3OH(g)$

t = 0	a	3a	0
t = t	$a - \frac{a}{4}$	$3a - \frac{a}{2}$	$\frac{a}{4}$
Partial Pressure	$\frac{3p}{14}$	$\frac{5p}{7}$	$\frac{p}{14}$

$K_P = \frac{p_{CH_3OH}}{p_{CO}\left(p_{H_2}\right)^2}$

$3 \times 10^{-4} = \frac{\frac{p}{14}}{\frac{3p}{14}\left(\frac{5p}{7}\right)^2}$

$p^2 = \frac{49 \times 10^4}{25 \times 3 \times 3}$

$p = \frac{7 \times 10^2}{5 \times 3} = 46.67 \text{ bar}$

(55) Answer : (2)

Solution:

at equilibrium Q = K_{eq}

(56) Answer : (3)

Hint:

At equilibrium K = Q

Solution:

$\Delta G = \Delta G^\circ + RT \ln Q$

At equilibrium, Q = K, $\Delta G = 0$

(57) Answer : (2)

Solution:

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$H_2 + I_2 \rightleftharpoons 2HI$

$K_{eq} \equiv \left(\frac{1}{8}\right)^2 = \frac{1}{64}$

(58) **Answer :** (1)
Hint:
If $K_C > 10^3$, then the reaction proceeds nearly to completion.

(59) **Answer :** (2)
Solution:
If $K_c < 10^{-3}$, reactants predominates over products.

(60) **Answer :** (4)
Hint:
Introduction of inert gas at constant volume and temperature does not affect the equilibrium state of a gaseous reaction.
Solution:
Partial pressure of reacting gases is unchanged on introduction of inert gas at constant volume hence there is no change in equilibrium state.
Introduction of inert gas at constant pressure and temperature shifts equilibrium towards more number of gaseous moles.

(61) **Answer :** (2)
Solution:
Catalyst does not affect the equilibrium composition of the reaction mixture.

(62) **Answer :** (1)
Hint:
At equilibrium $\Delta G = 0$
$\Delta G = \Delta G^\circ + RT \ln k_{eq}$
Solution:

	$2HI(g) \longrightarrow$	$H_2(g)$	+ $I_2(g)$
t = 0	1 mole	0	0
	1 – 0.5	0.25	0.25

$K_c = \frac{0.25 \times 0.25}{0.5 \times 0.5} = \frac{1}{4}$

$\Delta G^\circ = -2.303\ RT \log\left(\frac{1}{4}\right)$

= 1.386 RT

(63) **Answer :** (1)
Hint:
Apply Le Chatelier's principle.
Solution:
High temperature favours endothermic equilibrium, in forward direction.

(64) **Answer :** (2)
Solution:

$K_P = K_C(RT)^{\Delta ng}$

$\frac{K_C}{K_P} = \frac{1}{(RT)^{\Delta ng}}$

$\frac{K_C}{K_P} = \left(RT\right)^{-\Delta ng}$

a. $\Delta n_g = 2$

$\frac{K_c}{K_p} = (RT)^{-2}$

$\frac{K_c}{K_p} = \frac{1}{(RT)^2}$

b. $\Delta n_g = 0$

$\frac{K_c}{K_p} = \left(RT\right)^0 = 1$

c. $\Delta n_g = 1$

$\frac{K_c}{K_p} = \left(RT\right)^{-1} = \frac{1}{RT}$

d. $\Delta n_g = -1$

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$$\frac{K_c}{K_p} = \left(RT\right)^{-\Delta n_g}$$
$$= (RT)^1$$

(65) Answer : (3)
Solution:
Pressure increases, reaction shift in that direction where there is less number of gaseous moles.

(66) Answer : (3)
Hint:
K_C for a particular equilibrium change with temperature only.
Solution:

$$\log\left(\frac{K_2}{K_1}\right) = \frac{\Delta H}{2.303\,R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

If $T_2 > T_1$ (rise in temperature)
Then $K_2 > K_1$
If $\Delta H > 0$ (endothermic)
But $K_2 < K_1$
If $\Delta H < 0$ (exothermic)

(67) Answer : (1)
Solution:

Those substances which accept the H^+ are called conjugate base.

(68) Answer : (2)
Solution:

$$\alpha = \sqrt{\frac{K_a}{C}}$$
$$= \sqrt{\frac{1.8 \times 10^{-5}}{0.1}}$$
$= 1.34 \times 10^{-2}$
$= 1.34\%$

(69) Answer : (3)
Solution:

When $\left[\overline{O}\,H\right]_{SB} < 10^{-7}M$, contribution of OH^- from H_2O is also considered

$$H_2O \rightleftharpoons \underset{x}{H^+(aq)} + \underset{10^{-8}+x}{OH^-(aq)}$$

$$10^{-14} = x\left(10^{-8} + x\right)$$

Solving quadratic,
$[H^+] = x = 9.5 \times 10^{-8}$
pH = 7.02

(70) Answer : (2)
Hint:
$M_1V_1 + M_2V_2 = M_RV_R$
Solution:

$pH_1 = 2$ $\quad$ $pH_2 = 3$
$M_1 = [H^+]_1 = 10^{-2}$ $\quad$ $[H^+]_2 = 10^{-3} = M_2$
$V_1 = 100$ ml $\quad$ $V_2 = 100$ ml

$$M_R = \frac{10^{-2}\times 100 + 10^{-3}\times 100}{200} = 5\ .5\ \times 10^{-3}$$

$pH_{(Resultant\ solution)} = -\log[5.5 \times 10^{-3}]$
$pH_{(R)} = 2.26$

(71) Answer : (2)
Hint:
Lewis acids are electron pair acceptors.
Solution:

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B_2H_6 is an electron deficient species.

(72) **Answer :** (2)
Hint:
Electron deficient species are Lewis acids
Solution:
• Higher the bond polarity (H – A), stronger is the acidic strength.
• K_W of water at 298 K is 1×10^{-14}

(73) **Answer :** (1)
Solution:

$$K_c = \frac{K_f}{K_b} = \frac{8.5\times10^{-5}}{1.7\times10^{-4}} = 0.5$$

(74) **Answer :** (1)
Solution:
Salt of weak acid and strong base is basic in nature and thus have higher pH.

(75) **Answer :** (3)
Solution:

$$(CH_3COO)_2Ca \xrightarrow{H_2O} 2\,CH_3COO^- + Ca^{2+}$$

pH of salt of strong base and weak acid is

$$pH = 7 + \frac{1}{2}[pK_a + \log c]$$

$$= 7 + \frac{1}{2}[4.74 + \log 0.08]$$

= 7 + 1.83 = 8.82

(76) **Answer :** (3)
Solution:

(a) $\underset{10\,m\ eq}{NaOH} + \underset{10\ meq}{H_2SO_4} \rightarrow Na_2SO_4 + H_2O$

$\quad\quad -\quad\quad\quad -\quad\quad\quad\quad 10\ meq$

Na_2SO_4 do not undergo hydrolysis, so, $pH = 7$

(b) $\underset{10\,m\ mol}{NaOH} + \underset{10\ m\ mol}{CH_3COOH} \rightarrow \underset{-}{CH_3COONa} + H_2O$

$\quad\quad -\quad\quad\quad -\quad\quad\quad\quad 10\,m\ mol$

CH_3COONa undergoes anionic hydrolysis.

$$pH = 7 + \frac{1}{2}(pK_a + \log C)$$

(c) $\underset{10\,m\ mol}{NH_4OH} + \underset{10\,m\ mol}{CH_3COOH} \rightarrow \underset{-}{CH_3COONH_4} + H_2O$

$\quad\quad -\quad\quad\quad -\quad\quad\quad\quad 10\ m\ mol$

CH_3COONH_4 undergoes both cationic and anionic hydrolysis.

$$pH = 7 + \frac{1}{2}(pK_a - pK_b)$$

(d) $\underset{10\,m\ mol}{HCl} + \underset{10\,m\ mol}{NH_4OH} \rightarrow \underset{-}{NH_4Cl} + H_2O$

$\quad\quad -\quad\quad\quad -\quad\quad\quad\quad 10\ m\ mol$

NH_4Cl undergoes cationic hydrolysis

$$pH = 7 - \frac{1}{2}(pK_b + \log\ c)$$

(77) **Answer :** (1)
Hint:
100 mL of NaOH when added to CH_3COOH, creates acidic buffer solution.
Solution:

$$\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{10^{-5}}{10^{-1}}} = \sqrt{10^{-4}} = 10^{-2}$$

$[H^+] = c\alpha = 10^{-1} \times 10^{-2} = 10^{-3}$ M

pH = 3

b. $pH = pK_a + \log\frac{[Salt]}{[Acid]}$

[Salt] = [Acid]

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pH = pK_a = 5

c. At equivalence point salt hydrolysis will takes place

$pH = 7 + \frac{1}{2}(pK_a + \log C)$

Concentration of CH_3COO^- $= \frac{10 \text{ m moles}}{300 \text{ mL}}$

$pH = 7 + \frac{1}{2}\left(5 + \log\frac{1}{30}\right)$

$= 7 + \frac{1}{2}(5 - \log 30)$

= 8.76

(78) Answer : (1)

Solution:

	CH_3COOH	$\rightleftharpoons$	CH_3COO^-	+	H^+
Initial	0.02		–		–
Final	$0.02 - 0.02\alpha$		0.02α		$0.02\alpha + 0.2$

$K_a = \frac{0.02\alpha(0.02\alpha + 0.2)}{0.02(1-\alpha)}$

$K_a = \frac{0.2\alpha}{1}$

$\alpha = \frac{1.74 \times 10^{-5}}{0.2}$ = $8.7 \times 10^{-5} \times 100 = 8.7 \times 10^{-3} = 0.009\%$

(79) Answer : (4)

Hint:

Salt of weak acid or weak base will undergo hydrolysis

Solution:

Ammonium chloride (NH_4Cl) – Cationic hydrolysis

Sodium acetate (CH_3COONa) – Anionic hydrolysis

Ammonium acetate (CH_3COONH_4) – Both cationic and anionic hydrolysis

Sodium chloride (NaCl) – No hydrolysis, as it is a salt of strong acid and strong base

(80) Answer : (2)

Solution:

• Due to complex formation solubility of AgCl will increase in NH_3.

$AgCl + 2NH_3 \rightarrow [Ag(NH_3)_2]^+ Cl^-$

• Due to more common ion concentration of Cl^- (0.2 M), in $CaCl_2$ solubility of AgCl will be minimum.

(81) Answer : (1)

Solution:

Let the solubility of $Ca(OH)_2$ be s mol/L

$Ca(OH)_2 \rightleftharpoons \underset{s}{Ca^{2+}} + \underset{2s}{2OH^-}$

$K_{sp} = [Ca^{2+}][OH^-]^2$

$4 \times 10^{-6} = (s)(2s)^2$

$4 \times 10^{-6} = 4s^3$

$s^3 = 10^{-6}$

$s = 10^{-2}$

$[OH^-] = 2 \times 10^{-2}$

$pOH = -\log(2 \times 10^{-2})$

= 1.7

pH = 14 – 1.7

= 12.3

(82) Answer : (2)

Hint:

• $K_{sp(AgCl)} = [Ag^+][Cl^-]$

• In presence of common ion solubility decreases.

Solution:

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$AgCl \rightleftharpoons Ag^+ + Cl^-$

sM (s + 0.1)M [from NaCl]

$K_{sp} = [Ag^+]\,[Cl^-]$

$1.8 \times 10^{-10} = s \times 0.1 \quad (\because\ s + 0.1 \simeq 0.1)$

$s = 1.8 \times 10^{-9}$ M

(83) Answer : (3)
Solution:

$BaSO_4 : s^2 = K_{sp}$

$\therefore\ s = \sqrt{K_{sp}}$

$Hg_2Cl_2 : s \times (2s)^2 = K_{sp}$

$= 4s^3 = K_{sp}$

$s = \sqrt[3]{\frac{K_{sp}}{4}} = \left(\frac{K_{sp}}{4}\right)^{1/3}$

$Ca_3(PO_4)_2 : (3s)^3 \times (2s)^2 = K_{sp}$

$\Rightarrow 108\, s^5 = K_{sp}$

$s = \sqrt[5]{\frac{K_{sp}}{108}}$

$Fe(OH)_3 : s \times (3s)^3 = K_{sp}$

$\Rightarrow 27 s^4 = K_{sp}$

$\Rightarrow s = \sqrt[4]{\frac{K_{sp}}{27}}$

(84) Answer : (3)
Solution:

$K_w [H_3O^+]^2$

$= (10^{-6})^2$

$= 10^{-12}$

(85) Answer : (2)
Solution:
With change of pressure.
K_P remain constant.

$P(s) \rightleftharpoons 2Q(g) + 3R(g)$

$K_P = (P_Q)^2 (P_R)^3 = (P'_Q)^2\ (3P_R)^3$

$P_Q^2\ P_R^3 = P'_Q{}^2 \times 27 P_R^3$

$\frac{P_Q^2}{27} = P'_Q{}^2$

$P'_Q = \frac{P_Q}{3\sqrt{3}}$

(86) Answer : (4)
Solution:
α is almost one in strong electrolytes as they get completely dissociated.

(87) Answer : (4)
Solution:

$MX(s) \rightleftharpoons \underset{S}{M^+(aq)} + \underset{S}{X^-(aq)}$

For a saturated solution,

$K_{SP} = Q_{SP} = [M^+]\,[X^-]$

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$\Rightarrow 4 \times 10^{-8} = S.S \Rightarrow S = 2 \times 10^{-4}$

$MX_2(s) \rightleftharpoons M^{2+}(aq) + 2X^-(aq)$

$K_{SP} = S.(2S)^2 \Rightarrow 4S^3 = 3.2 \times 10^{-14}$

$\Rightarrow S = 2 \times 10^{-5}$

$M_3X(s) \rightleftharpoons M^+(aq) + 3X^-(aq)$

$K_{SP} = S.(3S)^3 \Rightarrow 27\ S^4 = 27 \times 10^{-16}$

$\Rightarrow S = 10^{-4}$

(88) Answer : (2)
Solution:

$$pOH = pK_b + \log \frac{[salt]}{[base]} = 5 + \log \frac{10}{1} = 6$$

(89) Answer : (3)
Solution:
On increasing volume and addition of inert gas at constant pressure equilibrium shifts in direction where number of moles of gaseous molecule increases.

(90) Answer : (1)
Solution:
Due to addition of KCN in aq solution of HCN, dissociation decreases due to common ion effect.

BOTANY

(91) Answer : (3)
Hint:
Protoxylem is formed towards periphery and metaxylem towards centre in roots.
Solution:
Vascular bundles are conjoint and closed in monocot leaf and monocot stem. Radial, exarch type of vascular bundles are found in roots. Pericycle in the form of semi-lunar patches are observed in dicot stem. Hypodermis is sclerenchymatous in monocot stem.

(92) Answer : (4)
Solution:
In monocot stem, vascular bundles are found scattered in whole ground tissue. The larger vascular bundles are loosely arranged towards the centre of the stem.

(93) Answer : (3)
Solution:
All the given features are true for dicot leaf. These leaves have palisade and spongy mesophyll tissues.

(94) Answer : (1)
Solution:
Conjunctive tissue is parenchyma cells which lie between xylem and phloem in dicot root. The conjoint vascular bundles usually have the phloem located only on the outer side of xylem.

(95) Answer : (4)
Solution:
Monocot stem does not have endodermis. Pericycle in monocot root has parenchymatous cells.

(96) Answer : (2)
Solution:
Both layers of epidermis are cuticularized.

(97) Answer : (3)
Solution:
Pith or medulla is present at centre of dicot stem.

(98) Answer : (3)
Hint:
Cell wall of root endodermal cells is impermeable to water because of the deposition of waxy substance.
Solution:
Waxy substance suberin is deposited in the form of casparian strips in root endodermal cells.

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(99) **Answer :** (2)
Solution:
The phloem parenchyma is absent in most of the monocots.

(100) **Answer :** (1)
Solution:
Pericycle is outer most layer of stele.
In monocots, pericycle forms lateral roots only. Monocots lack cambium.

(101) **Answer :** (1)
Solution:
The above given figure is of monocot stem.

(102) **Answer :** (2)
Solution:
Guard cells are parenchymatous.

(103) **Answer :** (2)
Solution:
In conjoint type of vascular bundles, the xylem and phloem are situated at the same radius of vascular bundles. Such vascular bundles are common in stems and leaves

(104) **Answer :** (1)
Solution:
In monocot root, pith is large and well developed.

(105) **Answer :** (3)
Solution:
Root epidermis is termed as epiblema.

(106) **Answer :** (1)
Solution:
When the bulliform cells in the leaves have absorbed water and are turgid, the leaf surface is exposed. When they are flaccid due to water stress, they make the leaves curl inwards to minimise water loss

(107) **Answer :** (1)
Solution:
Phloem parenchyma is absent in most of the monocots.

(108) **Answer :** (3)
Solution:
Root hair – Unicellular, unbranched
Trichome – Usually multicellular and branched/unbranched
Both are epidermal appendages.

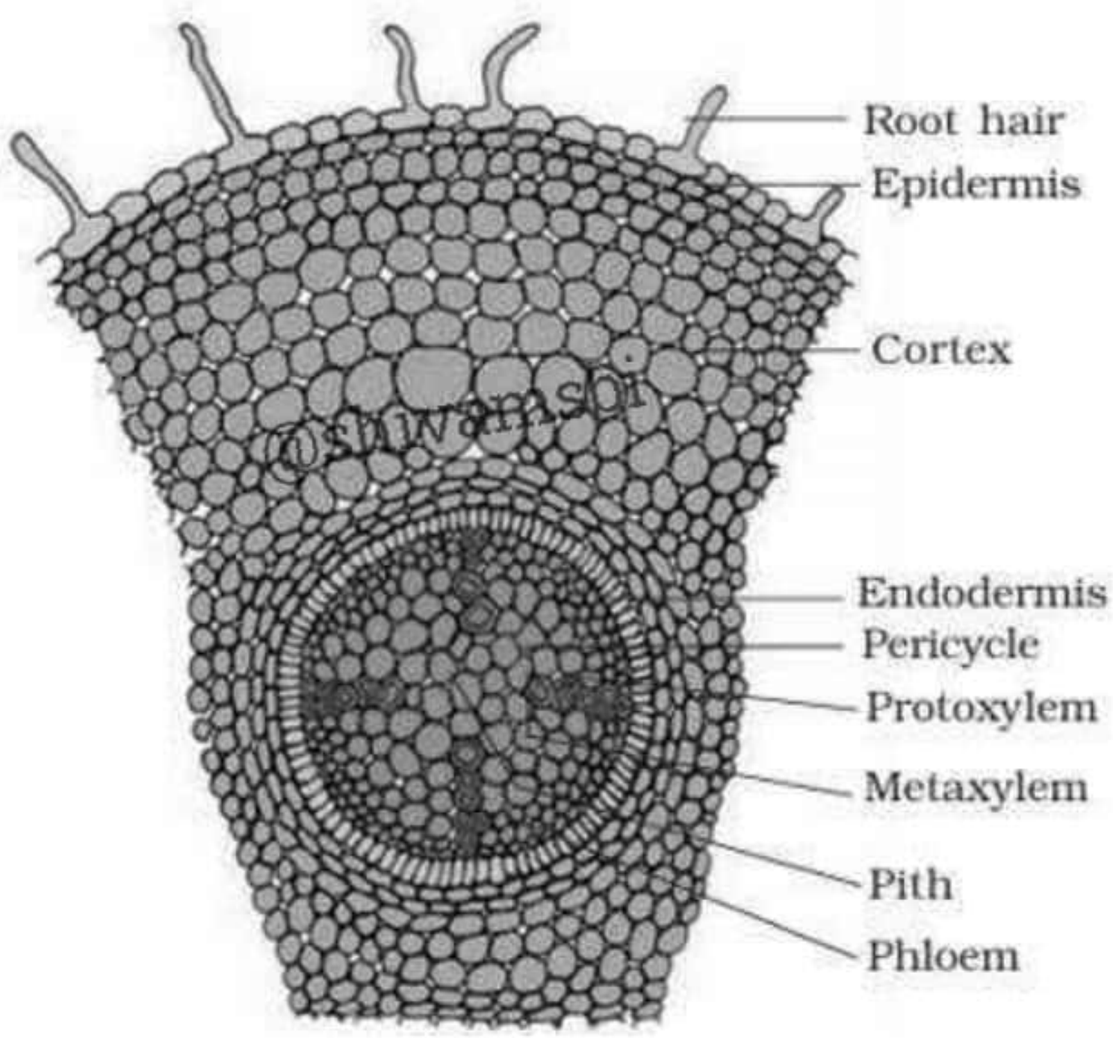
(109) **Answer :** (2)
Solution:
In roots, the vascular bundles are radial & exarch.

(110) **Answer :** (3)
Solution:
Trichome cannot be an organ of reproduction.

(111) **Answer :** (3)
Solution:

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(112) Answer : (3)
Solution:
All tissues on the inner side of the Endodermis constitute the stele.
Stele includes pericycle, vascular bundles and pith.

(113) Answer : (2)
Solution:
When the conjoint vascular bundle has cambium, it is called open vascular bundle.
These are present in stems of gymnosperms and dicot plants.

(114) Answer : (3)
Solution:
Bulliform cells are adaxial epidermal cells of monocot leaves which modify themselves into large, empty and colourless cells.
Companion cells are specialised parenchymatous cells of phloem.
Endodermal cells are cells of endodermis.

(115) Answer : (1)
Solution:
In stem and leaves of monocots and in dicot leaves, vascular bundles are conjoint and closed. In dicot stem,conjoint and open type of vascular bundle is found.

(116) Answer : (1)
Solution:
In dicot roots there are usually two to four xylem and phloem patches.

(117) Answer : (2)
Solution: Radially placed parenchymatous cells in between vascular bundles are known as medullary rays

(118) Answer : (3)
Solution:
The primary adaptive advantage provided by bulliform cells in grasses during water stress is that they reduce transpiration by causing leaf rolling to minimize exposed surface area.

(119) Answer : (2)
Solution:
The innermost layer of cortex is called endodermis.

(120) Answer : (2)
Solution:
In dicot stem, hypodermis is composed of 3-5 layers of collenchymatous cells.

(121) Answer : (3)
Solution:

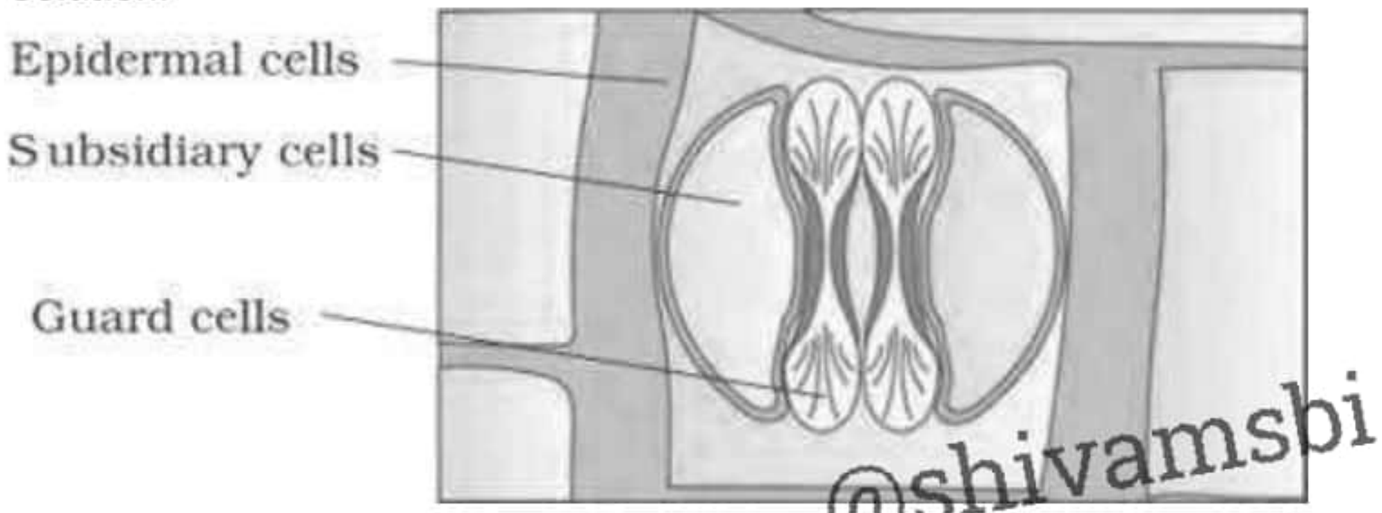
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Ground tissue of leaves is called mesophyll.

(122) Answer : (3)
Solution:
Xylem is a part of vascular tissue system and not the ground tissue system.

(123) Answer : (4)
Solution:
Stomatal apparatus consist of stomatal pore, guard cells and surrounding subsidiary cell.

(124) Answer : (3)
Solution:



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(125) Answer : (4)
Solution:
Grass leaves have dumb-bell shaped guard cells.

(126) Answer : (2)
Solution:
The given figure is of radial vascular bundles.
Radial vascular bundles are found in roots of both monocots and dicot roots.

(127) Answer : (1)
Solution:
In dicot stem, the vascular bundles are arranged in a ring.

(128) Answer : (2)
Solution:
The endodermis is also known as starch sheath in dicot stem

(129) Answer : (2)
Solution:
A waxy thick layer called cuticle which generally covers the epidermis and prevents water loss, is absent in roots.

(130) Answer : (2)
Solution:
Tissue is a group of cells having common origin, common or correlated function, structurally they can be similar or dissimilar, they can be meristematic or permanent.

(131) Answer : (3)
Solution:
Conjunctive tissue is present in the stele of dicot roots.

(132) Answer : (3)
Solution:
Trichomes are usually multicellular.

(133) Answer : (4)
Solution:
The innermost layer of the cortex is endodermis. Endodermis is rich in starch grains and it is known as starch sheath. Starch sheath is found in the dicot stem.

(134) Answer : (2)
Solution:
Roots possess radial vascular bundles while stem possess conjoint vascular bundles.

(135) Answer : (2)
Hint:
Vascular bundles of maize stem lack cambium.

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Solution:
In maize stem, bundle sheath is sclerenchymatous, vascular bundles are scattered in ground tissue and larger are towards the centre and smaller towards periphery.

ZOOLOGY

(136) Answer : (3)
Solution:
Osteoporosis in aged female after menopause is due to decreased levels of estrogen.

(137) Answer : (2)
Solution:

Joint between the carpals	–	Gliding joint
Joint between carpal and metacarpal of thumb	–	Saddle joint
Knee joint	–	Hinge joint
Joint between radius and carpal	–	Ellipsoidal joint

(138) Answer : (2)
Solution:
Each coxal bone of pelvic girdle is formed by fusion of three bones namely ilium, ischium and pubis.

(139) Answer : (1)
Hint:
It is not usually inheritable.
Solution:
In myasthenia gravis, destruction of acetylcholine receptors occurs due to auto-immune response. Muscular dystrophy is a genetic disorder leading to degeneration of skeletal muscles. Tetany occurs due to hypocalcemia.

(140) Answer : (4)
Solution:
Production of body heat is primarily related to adipose tissue and muscular activity.

(141) Answer : (1)
Solution:
Foetal stage - $C_7T_{12}L_5S_5C_4$
Adult stage - $C_7T_{12}L_5S_1$(fused)C_1(fused)

(142) Answer : (3)
Solution:
Each half of the pectoral girdle consists of a large, triangular bone called scapula and an elongated collar bone called clavicle.

(143) Answer : (3)
Hint:
8 carpals are arranged in two rows of one wrist.
Solution:
Cranial bones are 8 in number and the carpals of one wrist are also 8 in number. An adult human possesses 14 facial bones, 26 serially arranged vertebrae in vertebral column, 22 skull bones and 7 tarsals in single hindlimb.

(144) Answer : (1)
Solution:
Palatine bone makes posterior part of hard palate and forms floor and lateral wall of nasal cavity.
Maxillae articulates with every bone of face, except mandible. Hyoid is present at the base of buccal cavity.

(145) Answer : (4)
Solution:
Each half of the pectoral gridle consists of two bones – clavicle and scapula.

(146) Answer : (3)
Solution:
Tetany is characterised by rapid spasms (wild contractions) in muscles due to low Ca^{++} in body fluids.
Parathyroid hormone (PTH) increases Ca^{+2} levels in blood, so it is a hypercalcemic hormone.

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(147) Answer : (1)
Hint:
No movement is allowed
Solution:
Fibrous joint is present between the flat skull bones which are fused end to end with the help of dense fibrous connective tissue in the form of sutures to form the cranium.
Cartilaginous joint is present between the adjacent vertebrae in the vertebral column.
Ball and socket joint, pivot joint and hinge joint are examples of synovial joints.

(148) Answer : (4)
Hint:
Longest bone in the human body
Solution:
Pelvic girdle consists of two coxal bones. Each coxal bone is formed by the fusion of three bones – ilium, ischium and pubis. At the point of the fusion of above bones is a cavity called acetabulum to which the thigh bone articulates. The two halves of the pelvic girdle meet ventrally to form the pubic symphysis.

(149) Answer : (1)
Solution:
Scapula has a shallow articular surface called glenoid cavity to which humerus articulates to form the shoulder joint.

(150) Answer : (2)
Hint:
Multiple of 3
Solution:
The bones of the limbs along with the girdles constitute the appendicular skeleton. Each limb is made of 30 bones.

(151) Answer : (3)
Solution:
Bone marked as 'X' is sphenoid bone which forms the base of the cranium. It is not a facial bone.

(152) Answer : (2)
Solution:
8th, 9th and 10th pairs of ribs are not directly connected with sternum but they join the 7th rib with the help of hyaline cartilage. So, these ribs are called as vertebrochondral ribs.

(153) Answer : (3)
Solution:
Metacarpals are included in bones of forelimbs.

(154) Answer : (1)
Solution:
Hyoid is a single U-shaped bone that does not articulate with any other bone.
Mandible = Lower jaw = Floor of buccal cavity

(155) Answer : (2)
Solution:

First seven pairs of ribs (14 in number)	–	True ribs
16 in number	–	Carpals (Wrist bones)
14 in number	–	Facial bones
Last 2 pairs of ribs (4 in number)	–	Floating ribs
3 pairs of ribs (6 in number)	–	False ribs

(156) Answer : (1)
Solution:
The PNS comprises of nerves, nerve fibres, ganglia and plexuses.

(157) Answer : (3)
Solution:
Unipolar neurons are present in embryonic stage and have single process.

(158) Answer : (3)
Hint:
This system involves neurotransmitters.
Solution:

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Chemicals called neurotransmitters are released at synapses between neurons to facilitate signalling. Endocrine system requires hormones to provide chemical integration.

(159) Answer : (4)
Solution:
Hint: Stimulus can be external or internal.
Sol. : Neurons are responsible for detection of stimulus (*via* neuronal receptors, proprioceptors, *etc*.). They receive stimulus and can transmit different types of impulses from one place to another.

(160) Answer : (1)
Solution:
The autonomic neural system is further classified into sympathetic neural system and parasympathetic neural system.

(161) Answer : (4)
Hint:
Hypothalamus lies at the base of thalamus.
Solution:
The medulla oblongata contains centres which control respiration, cardiovascular reflexes and gastric secretions.
Thermoregulatory centre is present in the hypothalamus.

(162) Answer : (1)
Solution:
At rest, the axonal membrane is nearly impermeable to Na^+.

(163) Answer : (2)
Hint:
Arbor vitae.
Solution:
Cerebellum is a part of hindbrain and is not a structural component of the brain stem.

(164) Answer : (2)
Hint:
A neuron has a cell membrane.
Solution:
Axolemma is the term used for neuronal cell membrane in axon region. Schwann cells form the myelin sheath around neurons of PNS. Oligodendrocytes form the myelin sheath around the neurons of CNS.

(165) Answer : (3)
Hint:
Connexons/Gap junctions play a key part in these synapses.
Solution:
Electrical synapses are effective due to the presence of gap junctions. They do not involve the use of neurotransmitters and permit the bidirectional flow of ions between cells.

(166) Answer : (3)
Hint:
Perikaryon is a part of this cell.
Solution:
Nerve cells have cyton, dendrites, axon and nodes of Ranvier as characteristic structures.

(167) Answer : (3)
Hint:
Possess Nissl's granules
Solution:
The cerebral cortex is referred to as the grey matter due to its greyish appearance. The neuron cell bodies are concentrated here giving the colour.
Fibres of the tracts are covered with the myelin sheath, which constitute the inner part of cerebral hemisphere. They give an opaque white appearance to the layer and hence, is called the white matter.

(168) Answer : (1)
Solution:
Both cerebral hemispheres are connected by a tract of nerve fibres called corpus callosum.

(169) Answer : (1)
Solution:
Associated deep structures like amygdala, hippocampus, *etc*., form a complex structure called the limbic lobe/limbic system.

(170) Answer : (3)
Hint:

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Layer whose latin translates into soft mother.
Solution:
Dura mater is the outer layer of the cranial meninges. It is thick and tough layer made up of tough fibrous tissue. Arachnoid is very thin middle layer made up of delicate connective tissue.
The pia mater is the innermost layer of the meninges, which directly adheres to the surface of the brain and spinal cord, and is highly vascularized to supply blood to the brain tissue.

(171) **Answer :** (3)
Hint:
Identify the cnidarian
Solution:
Hydra possesses primitive type of neurons. It has nerve cells which form an irregular nerve net but no brain.

(172) **Answer :** (1)
Hint:
Area which is neither clearly sensory nor motor in function.
Solution:
The cerebral cortex contains three major types of areas *i.e.*, motor, sensory and association areas. Sensory area receives various types of sensations. Motor areas are concerned with movements of skeletal elements. Association areas are responsible for certain complex functions like intersensory association, memory and communication.

(173) **Answer :** (3)
Solution:
Hint : Centre which regulates respiratory rhythm centre.
Sol. : Pneumotaxic centre is situated in pons which is a part of hind brain whereas hunger, thirst and thermoregulatory centres are situated in the hypothalamus which is a part of prosencephalon (forebrain).

(174) **Answer :** (4)
Solution:
Limbic system, along with hypothalamus, is involved in the regulation of sexual behaviour, expression of emotional reactions (*eg*., excitement, pleasure, rage and fear)

(175) **Answer :** (2)
Solution:
Cerebral aqueduct is a canal which passes through midbrain and connects 3rd ventricle of diencephalon with 4th ventricle of medulla oblongata.

(176) **Answer :** (1)
Hint:
Neuron with one axon and two or more dendrites.
Solution:
Based on the number of axon and dendrites, the neurons are divided into three types, *i.e.,* multipolar (with one axon and two or more dendrites; found in the cerebral cortex), bipolar (with one axon and one dendrite; found in the retina of eyes) and unipolar (cell body with one axon only; found usually in the embryonic stage).

(177) **Answer :** (2)
Solution:
The mid brain is located between the thalamus/hypothalamus of the forebrain and pons of the hindbrain.

(178) **Answer :** (4)
Hint:
Constituent of brain stem
Solution:
The medulla oblongata contains centres which control respiration, cardiovascular reflexes and gastric secretions. Pons, cerebellum and medulla are parts of hindbrain. Cerebrum is a part of forebrain.

(179) **Answer :** (2)
Hint:
Present at axon terminals
Solution:
The branch of axon terminates as a bulb-like structure called synaptic knob which possesses synaptic vesicles containing chemicals called neurotransmitters. The gaps between two adjacent myelin sheaths are called nodes of Ranvier. A synapse is formed by the membranes of a pre-synaptic neuron and a post-synaptic neuron, which may or may not be separated by a gap called synaptic cleft.

(180) **Answer :** (3)
Hint:
A gap may or may not be present in a synapse.
Solution:

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A synapse is formed by the membranes of a pre-synaptic neuron and a post-synaptic neuron, which may or may not be separated by a gap called synaptic cleft.
There are two types of synapses, namely electrical synapses and chemical synapses. At electrical synapses, the membranes of pre and post-synaptic neurons are in very close proximity. Electrical current can flow directly from one neuron into the other across these synapses.
At a chemical synapse, the membranes of the pre-synaptic and post-synaptic neurons are separated by a fluid-filled space called synaptic cleft.

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